The Economics of Pet Ownership: A Quantitative Analysis

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SUMMARY

This paper develops a family of microeconomic models of pet ownership to analyze how households allocate time and financial resources across pet ownership and care. The models treat the household as a utility-maximizing agent subject to joint budget-time constraints that capture both monetary pet expenditures and the opportunity cost of caregiving time. Analytical results characterize optimal pet quantity, care intensity, and consumption as functions of exogenous parameters. Comparative-statics analysis yields clear, testable hypotheses about parameter effects on household choices, and their robustness is evaluated through alternative models. Computational experiments visualize the utility landscape and illustrate how key parameters shape optimal pet ownership, care, and expenditures.

KEYWORDS

Pet ownership, household decision-making, optimization, comparative statics, time allocation

INTRODUCTION

Pet ownership is widespread. In 2024, 71% of U.S. households (94 million) owned pets and spent $152 billion on them[1]. Pet owners devoted 44 minutes a day to pet care—nearly as much as grocery shopping (46 min) and food preparation (53 min)[2]. Moreover, 97% of pet owners consider pets part of the family[3].

Despite the increasingly central role of pets in households, they remain largely absent from economic analyses of the family. Therefore, this work develops a set of microeconomic models of pet ownership to analyze how households allocate time and financial resources when making pet ownership and care decisions. Assuming demand for pets is analogous to demand for children, the proposed models build on economic theories of fertility, following Becker[4] and subsequent advances[5].

Few economic models formally examine household choices about pet ownership, except Schwarz et al.[6], which analyzes household income, composition, and pet expenditures to estimate the income elasticity of pet spending. In contrast, the present work develops a multi-model framework of pet ownership that explicitly incorporates caregiving time, its opportunity cost, and joint budget-time constraints under utility maximization. Such time-allocation mechanisms are not considered in Schwarz et al., even though time constraints are among the primary reasons for pet relinquishment[7].

METHODS

The proposed framework is a family of models that define the household's optimization problem (Table 1). Each model treats the household as a rational economic agent that maximizes utility by choosing pet quantity n, caregiving intensity c, and consumption s. The utility function consists of s and companionship from pets ($\alpha \ln(\beta n q)$ in the baseline model; M1), where the logarithmic term implies diminishing marginal returns. In M1, the household solves $\max U(s, n, q) = s + \alpha \ln(\beta n q)$ subject to the joint budget-time constraint. The companionship term depends on n, pet welfare q, and the household's preference for pets $\alpha > 0$, while $\beta > 0$ scales the product $n \cdot q$. Pet welfare is produced through caregiving intensity c, which captures the quality and resource intensity of care, such as food quality and exercise intensity. The exponent γ ($0 < \gamma < 1$) governs how c is converted into pet welfare q.

Each model incorporates a budget constraint and a time constraint (Table 1). The budget constraint allocates income w between consumption s and pet expenditures $(pc + k)n$, where p is the unit cost of pet care (e.g., food and hygiene items) and k is the fixed cost per pet (e.g., adoption fees). The time constraint allocates total available time between labor supply and pet care, where caregiving time per pet t reduces income through its opportunity cost wt. Combining these constraints yields a joint budget-time constraint that allocates income across s, $(pc + k)n$, and wtn (Table 1).

The household's optimization problem is solved using first-order conditions. The marginal condition for

Table 1 A Family of Household Utility Models of Pet Ownership

Model	Utility Function	Budget-time Constraint
Baseline (M1)	$U(s,n,q) = s + \alpha \ln(\beta n q),\quad q = c^{\gamma}$	$w = s + (pc + wt + k)n$
Model 2 (M2)	$U(s,n,q) = s + \alpha \ln(\beta n q),\quad q = c^{\gamma} t^{1-\gamma}$	$w = s + (pc + wt + k)n$
Model 3 (M3)	$U(s,n,q) = s + \alpha \ln(\beta n^{\epsilon} q),\quad q = c^{\gamma} t^{1-\gamma}$	$w = s + (pc + wt + k)n$
Model 4 (M4)	$U(s,n,q) = s + \alpha_n \ln(n) + \alpha_q \ln(q),\ q = c^{\gamma}$	$w = s + (pc + wt + k)n$
Model 5 (M5)	$U(s,n,q) = s + \alpha_n \ln(n) + \alpha_q \ln(q),\ q = c^{\gamma} t^{1-\gamma}$	$w = s + (pc + wt + k)n$
Model 6 (M6)	$U(s,n,c,t_d) = s + \alpha \ln(\beta n^{\epsilon} q),\quad q = c^{\gamma} t_d^{\delta}$	$w = s + (pc + w(t_r + t_d) + k)n$

Table 2 Comparative-statics Signs, Robustness, and Calibrated Elasticities for the Baseline Model

	Comparative Statics Signs						Elasticities					
	α	γ	w	p	k	t	α	γ	w	p	k	t
Pet quantity n^*	+ ~	− ✓	− ✓	0 ✓	− ✓	− ~	1	-0.18	-0.95	0	-0.05	-0.95
Care intensity c^*	0 ~	+ ✓	+ ✓	− ✓	+ ✓	+ ~	0	1.18	0.95	-1	0.05	0.95
Standard of living s^*	− ~	0 ✓	+ ✓	0 ✓	0 ✓	0 ✓	-0.07	0	1.07	0	0	0
Pet expenditure $pc^*n^* + kn^*$	+ ~	+ ✓	− ✓	0 ✓	+ ✓	− ~	1	0.74	-0.21	0	0.04	-6.47
Total care time tn^*	+ ~	− ✓	− ✓	0 ✓	− ✓	+ ~	1	-0.18	-0.95	0	-0.05	0.05

pet quantity n equates the marginal benefit of an additional pet with its marginal cost, where the cost includes both monetary expenditures and the opportunity cost of caregiving time. Similarly, the marginal condition for caregiving intensity c equates the marginal benefit of improving pet welfare with the unit cost of care. These conditions determine optimal n^* and c^* as functions of exogenous parameters.

Beyond M1, M2–M6 vary the companionship term and pet-welfare function to assess the robustness of results across alternative models while preserving the same optimization structure. Full details on model formulations, derivations, and first- and second-order conditions are provided on the project webpage[8].

RESULTS AND DISCUSSION

From the optimality conditions, comparative-statics analysis of the baseline model yields hypotheses about how exogenous parameters (columns in Table 2; e.g., α and γ) affect household choices (rows in Table 2). "+", "−", and "0" indicate whether an increase in a parameter raises, lowers, or does not affect the corresponding outcome. ✓ and ~ indicate that these hypotheses are robust or partially robust across the six models. Table 2 shows that these qualitative predictions remain largely stable.

These hypotheses help explain recent trends in pet ownership and expenditures (Figure 1)[1]. During the pandemic, increased time at home reduced the opportunity cost of caregiving time wt, which the model predicts would increase optimal pet ownership n^* (Table 2) . In contrast, as economic activity resumed and time constraints tightened, the opportunity cost increased, contributing to a decline in pet ownership. Meanwhile, pet expenditures ($pc^*n^* + kn^*$) may continue to rise because of Gen Z-driven stronger preferences for pets α, improved caregiving efficiency γ from the "pet tech" boom (e.g., automatic food dispensers and wearable activity trackers), and inflation-driven increases in fixed ownership costs k.

Figure 2 visualizes the household utility landscape through computational experiments calibrated to empirical U.S. datasets, such as the U.S. Census, ATUS[2], and APPA[1]. In the 2D landscape, the red dot represents the household's optimal pet quantity and caregiving intensity (n^*, c^*). Colored curves are utility contours (indifference curves), each of which represents combinations of n and c that yield the same level of utility. The black curve shows the set of feasible (n, c) combinations under the allocation of monetary expenditures ($pcn + kn$) and opportunity cost (wtn) at the optimal consumption s^*. This figure highlights the tradeoff structure: the curves are downward sloping in most regions, indicating that an increase in n must be offset by a reduction in c to maintain the same level of utility.

Calibrating comparative-statics results to the empirical datasets, Table 2 presents the elasticities of household choices with respect to exogenous parameters. The most notable is the elasticity of pet expenditure with respect to caregiving time t; 1% increase in t reduces optimal pet expenditure by 6.47%, as households respond to tighter time constraints by reducing ownership-related spending.

Figure 3 illustrates the marginal benefit-cost condition for pet quantity. The marginal benefit (MB) curves slope downward due to diminishing returns to companionship, while the marginal cost (MC) curves

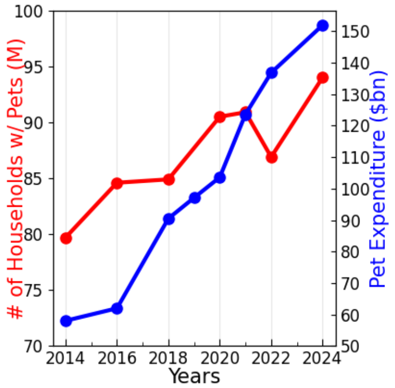

Figure 1 Pet ownership trend

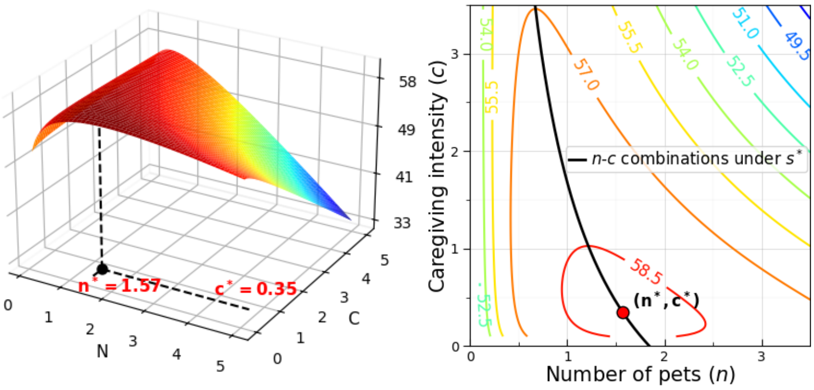

Figure 2 Utility landscape and optimal choice (3D: left; 2D: right)

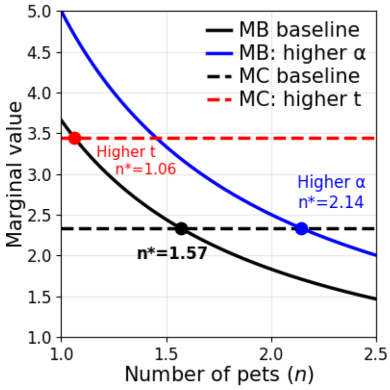

Figure 3 Marginal benefit-cost analysis

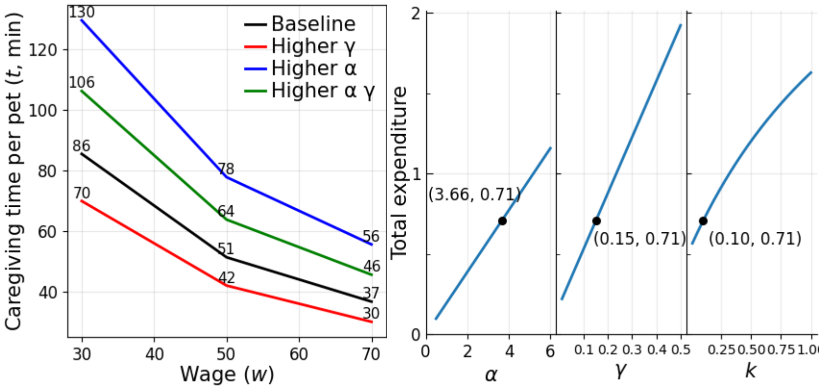

Figure 4 Comparative statics of optimal choices

slope upward due to monetary costs and the opportunity cost of caregiving time. Their intersection determines n^*. Changes in parameters (e.g., α and t) shift the MB and MC curves and alter n^*.

Figure 4 summarizes comparative statics. Its left panel illustrates the relationship between wage w and caregiving time t when $n^* = 1$. It shows that t falls with w because higher w raise the opportunity cost of time, although higher α and lower γ partially offset this decline. The other three panels show that total pet expenditure rises with α, γ, and k, with calibrated parameter values marked on each curve.

CONCLUSIONS

This paper develops a family of models of pet ownership that integrates monetary and time costs within a unified optimization problem. A key finding is that time costs play a central role in pet-related household choices and help explain U.S. trends in pet ownership and expenditures. Comparative-statics predictions are largely robust across model variants, and empirical calibration indicates that they are quantitatively meaningful, particularly for household responses to caregiving time requirements. Future work includes regression analysis to test comparative-statics results with empirical datasets.

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